Final Presentation Script (Revised)

DOSM Datathon 2026 Final · Team Nexora (UTAR) · How does geopolitical risk affect tourist arrivals to Malaysia? · Final deck, 12 slides + 4 backups. This is a new version; the original script is *pitch_script_final.pdf*.

#	Slide	Speaker	Time	Length
1	Cover	A	0:00 – 0:25	25 s
2	Tourism matters to Malaysia	A	0:25 – 1:10	45 s
3	Test every market, then forecast ahead	A	1:10 – 1:35	25 s
4	Four steps, one rule	A	1:35 – 2:45	1 min 10 s
5	The answer is both	B	2:45 – 3:20	35 s
6	Drops and rises cancel out	B	3:20 – 3:45	25 s
7	LightGBM had the smallest error	B	3:45 – 4:10	25 s
8	Most accurate 1–2 months ahead	B	4:10 – 4:40	30 s
9	Live demo: four questions	B	4:40 – 7:00	2 min 20 s
10	Useful, on-theme and realistic	A	7:00 – 8:05	1 min 05 s
11	Three things we found	A	8:05 – 8:45	40 s
12	Close	A	8:45 – 9:15	30 s
	Total		**about 9:15**	**45 s spare**

Times assume a calm pace of about 130 words a minute, plus a few seconds for each click. The talk is planned for **about 9:15**, leaving 45 seconds spare for handovers, switching to Power BI, a slow or frozen demo and nerves. **Rehearse with a timer.** If you are running late, cut in this order: (1) the Feature Importance line in the demo, (2) the Singapore and South Korea clicks in the demo, (3) the SDG sentences on slide 10. Never cut the findings or the limits.
Orange italic lines are actions, not words to say. ▶ *Click* shows the next item on the same slide; ▶ *Next slide* moves on. The click cues match the final deck. **Bold words** are the ones to stress. Speaker A presents slides 1–4 (problem and method) and 10–12; Speaker B presents slides 5–9 (findings and demo). In Q&A, all four members answer (see the last pages).

The story thread. Slide 1 asks about 2022, when global risk nearly doubled. The demo shows the answer: in 2022, arrivals still jumped, because borders reopened, so the national total hides the link. Slide 12 closes the loop: "back to 2022". Keep these three moments sharp; they tie the talk together.
Delivery. Pause one beat after each key number (+0.08, 16 of 20, 16 thousand). Look at the judges, not the screen. Speaker B: practise the demo clicks at least five times on the presentation laptop. Better still, let a third member drive the mouse while Speaker B narrates and faces the judges; it shows teamwork and keeps eye contact. In the demo, follow the same pattern on every page: ask the page's question, click, then say the takeaway out loud. In Q&A: answer in two or three sentences, then stop; if you don't know, say so and say what you would check.

Slide 1 · Cover

Speaker A · 0:00 – 0:25 · PJT1 · INV3 · INV4

Good morning, judges. In 2022, Russia invaded Ukraine, and world risk nearly doubled. So, did fewer tourists come to Malaysia?

That is our question: how does geopolitical risk, meaning wars, conflicts and political tension, affect tourist arrivals to Malaysia? We are Team Nexora, four students from UTAR, with our advisor Dr. Kelwin Tan. We studied Malaysia's top 20 source markets, the countries our tourists come from.

▶ *Next slide*

Slide 2 · Tourism matters to Malaysia

Speaker A · 0:25 – 1:10 · PJT2 · IMP2 · ANL1

Tourism is 15.9 percent of our economy and gives 3.7 million jobs. So when the world becomes risky, it can hit our hotels, airlines and homestays.

▶ *Click* But here is the problem. If we look at Malaysia as one total, the link between risk and tourists is only plus 0.08. **Almost zero.** It looks like risk does not matter.

▶ *Click* Lee and others found the same in 2021. And nobody had checked each country one by one. So planners cannot tell if a drop comes from every country or only a few. That is the gap we study.

▶ *Next slide*

Slide 3 · Test every market, then forecast ahead

Speaker A · 1:10 – 1:35 · PJT2 · PJT3 · INV3

So we have two goals. One: test how risk links to tourists from each of our top 20 markets. Two: forecast tourist arrivals one to four months ahead.

Now, a quick question. When risk goes up, do tourists come *(click A)* less, *(click B)* more, or *(click C)* both? *Pause for two seconds and look at the judges.*

Here is how we found out.

▶ *Next slide*

Slide 4 · Four steps, one rule

Speaker A · 1:35 – 2:45 · PJT4 · ANL2 · ANL4 · INV1 · INV5

We worked in four steps.

▶ *Click* Collect. Only official, open data: tourist arrivals, exchange rates, fuel and oil prices, and a world risk index. About 69 thousand daily rows became 2,300 monthly records.

▶ *Click* Date. Each number is dated by the day it is published, not the day it happened. Without this, a model looks better in testing than in real life.

▶ *Click* Model. Two models, two jobs. Ridge, a simple straight-line model, gives one clear risk number per country. LightGBM, built from many small decision trees, makes the forecast.

▶ *Click* Test. We tested five past dates as if live, plus a COVID stress test.

▶ *Click* And one rule: the model only learns from numbers already published. **It never sees the future.**

My teammate will now share what we found.

▶ *Next slide · hand over to Speaker B*

Slide 5 · The answer is both

Speaker B · 2:45 – 3:20 · ANL1 · ANL3 · PJT4

Thank you. Our first finding answers goal one. Less, more, or both? The answer is **C: both.**

We tested each country, keeping oil price, exchange rate, season and past tourists the same. We found three groups.

▶ *Click* For **16 of our 20 markets**, like Nepal, South Korea and Germany, higher risk goes with fewer tourists.

▶ *Click* For 3 markets, Singapore, Indonesia and China, higher risk goes with more tourists.

▶ *Click* And for Thailand, there is almost no change.

▶ *Next slide*

Slide 6 · Drops and rises cancel out

Speaker B · 3:20 – 3:45 · ANL5 · ANL3 · INV2

This is our key idea. When we add all countries together, the drops and the rises cancel out, so the total looks flat. This may explain why past research found no link.

▶ *Click* So we must watch risk **market by market**. But remember, these are patterns, not proof that risk causes the change.

▶ *Next slide*

Slide 7 · LightGBM had the smallest error

Speaker B · 3:45 – 4:10 · ANL2 · ANL3 · ANL4

Our second finding answers goal two: forecasting. A model must beat simple rules, so we tested it on past data as if it were live. LightGBM had the smallest error, about 16 thousand tourists per country per month. Ridge came close. LightGBM is about 7 percent better than repeating last month. A fair win, not a big one.

▶ *Next slide*

Slide 8 · Most accurate 1–2 months ahead

Speaker B · 4:10 – 4:40 · ANL3 · ANL5

Our forecasts are best **one and two months ahead**, with error under 12 thousand. At three and four months, the error is about double.

▶ *Click* One honest result: during COVID, simply repeating last month beat our model, because a shock like that was not in the past data. So in a sudden shock, trust the latest month.

▶ *Next slide*

Slide 9 · Live demo: four questions

Speaker B · 4:40 – 7:00 · DEM1–5 · INV4

Now our dashboard. It answers four questions, one page each, using the buttons on the left. We start on the Overview, which covers our 20 markets.

Alt+Tab to Power BI. It should already be open on Malaysia Tourism Overview, full screen, both slicers set to All. Before the day, check that South Korea is in the Source Country list on Tourism Drivers, and find the 2022 point on the red line of the Tourist Arrivals vs Global Geopolitical Risk chart so you can point at it without searching. If a judge points at the 16.8% card, use the Q&A answer.

1. Which markets matter? (25 s). *Click Market Analysis.* This page helps plan promotion. ASEAN brings almost three in four of our tourists. *Click the Non-ASEAN part of the donut.* Outside ASEAN, China leads. But big does not mean sensitive to risk. *Click blank space to clear.*

2. What moves arrivals? (45 s). *Click Tourism Drivers. Point at the last blue bar.* This last bar is only January to May 2026, so it is not a drop. *Point at Tourist Arrivals vs Global Geopolitical Risk, then at 2022 on the red line.* Remember 2022? World risk nearly doubled, **but tourists went up**, because borders reopened after COVID. Added together, the recovery hides the link. *Set Source Country to*

Singapore. So we look at one country at a time. Here is Singapore. *Set Source Country to South Korea.* And here is South Korea. Each has its own pattern, so we test every country separately. *Set Source Country back to All.*

3. What do we forecast? (30 s). *Click Forecast and point at Forecast Arrivals 2026.* The model forecasts about 2.3 million tourists a month in June and July 2026, then about 2.1 million by September. **So:** a steady June and July, then a normal seasonal dip. *Set source_country_name to Singapore, then back to All. Point at Feature Importance.* The model relies most on recent tourists and the season. World risk is sixth.

4. How accurate is it? (20 s). *Point at Model Accuracy | Avg MAE (K).* LightGBM's average error is about 16 thousand per country per month. *Point at COVID Stress Test.* During COVID, repeating last month won, and we show that openly.

So, four pages: where tourists come from, what moves them, what is coming, and how much to trust it.

Alt+Tab back to the slides. If Power BI freezes, click Demo backup and talk through the screenshots instead.

▶ *Next slide · hand over to Speaker A*

Slide 10 · Useful, on-theme and realistic

Speaker A · 7:00 – 8:05 · IMP1–5 · INV5

So why does this matter?

▶ *Click* First, it is useful for three groups. Tourism Malaysia can see which countries send fewer tourists when risk rises, and promote there. DOSM can add a country view next to the national total. And hotels, airlines and homestays can plan staff one to two months ahead.

▶ *Click* Second, it fits this year's theme, machine learning for sustainable tourism. Better planning supports steady tourism jobs, which is SDG 8.9. The dashboard is a tool to monitor tourism, which is SDG 12.b. It also supports the National Tourism Policy 2020 to 2030.

▶ *Click* Third, it is realistic. It uses open, official data, updates every month with one command, and can grow to more countries and states.

To be honest, these are possible uses. We have not measured the real impact yet.

▶ *Next slide*

Slide 11 · Three things we found

Speaker A · 8:05 – 8:45 · ANL5 · PJT2 · PJT3

To sum up, we found three things. First, each market reacts differently to risk.

▶ *Click* Second, looking market by market shows what one total number hides.

▶ *Click* Third, our forecasts are best one to two months ahead.

▶ *Click* We are also honest about our limits. These are patterns, not causes. Sudden shocks like COVID cannot be predicted. And we only cover 20 markets. Next, we will add forecast ranges and re-check the three rising countries with recent years.

▶ *Next slide*

Slide 12 · Close

Speaker A · 8:45 – 9:15 · PJT1 · PJT3 · INV2 · INV3; Q&A: PJT5

So, back to 2022. Did fewer tourists come to Malaysia? In total, no. Tourists went up, because borders reopened. But for **16 of our 20 markets**, higher risk goes with fewer tourists, though not for all. That is why we look **market by market**.

▶ *Click* Terima kasih. We are happy to take your questions.

Q&A: click Accuracy, Markets, Data & design or Demo on this slide to open backups 13–16; each backup has a Q&A index button back to slide 12.

Marking scheme coverage

Every official criterion, and where the script earns it. Weights: C1 PJT 20% · C2 DEM 25% · C3 ANL 20% · C4 IMP 25% · C5 INV 10%. Slide headers carry the same codes.

Code	Criterion (short)	Where the script earns it
PJT1	Clear, organised, easy to understand	One idea per slide; plain English; jargon explained (Ridge = straight-line model, LightGBM = decision trees)
PJT2	Mastery of problem, objectives, solution	Slide 2 problem (national number hides markets; planners can't tell broad vs concentrated drops); slide 3 two objectives; slides 5 and 7 answer them by name
PJT3	Logical flow, keeps to time	Question → gap → objectives → method → findings → demo → impact → conclusion; "objective one/two" signposts; 2022 thread from slide 1 to 12; about 9:15 with 45 s spare
PJT4	Methods and results explained confidently	Slide 4 four steps with a reason for each; delivery notes (stress words, pauses)
PJT5	Answers questions accurately, with evidence	Q&A sheet: 24 answers with numbers and backup slides; owner for each question
DEM1	Demo runs smoothly	Prep checklist (start page, slicers, South Korea, 2022 point); click practice; third member drives; Demo backup slide
DEM2	Clear navigation, interactivity, UX	Page buttons named; donut cross-filter; Source Country slicer; single-market forecast
DEM3	Visuals support main findings	2022 on the risk chart shows why the total hides the link; Singapore vs South Korea shows market-by-market
DEM4	Neat, professional, real-world layout	Reading notes on every chart mentioned; "2026 so far" bar explained so it is not misread
DEM5	Relevant functions, value for target users	Each page tied to a use (promotion planning; hotel/airline planning; how far to trust it); takeaway said on each page; four-page summary
ANL1	Accurate, relevant, aligned with problem	Findings answer the stated objectives; scope stated (20 markets)
ANL2	Appropriate methods and models	Ridge for the link, LightGBM for the forecast; rolling backtests; COVID stress test; simple-rule baselines
ANL3	Findings supported by evidence	16/3/1 with controls named; error 16 thousand, 7% better than repeating last month; under 12 thousand at 1–2 months
ANL4	Rationale for analytical decisions	Why publish dates (testing would look better than real life); why two models; why compare with simple rules
ANL5	Critical thinking, meaningful insight	Opposite markets cancel out, which may explain past research; COVID result shown openly; limits and next steps on slide 11
IMP1	Real-world potential (government, industry, society)	Tourism Malaysia and DOSM (government); hotels and airlines (industry); local homestays (society)
IMP2	Solves a real social, economic or operational issue	Slide 2 states the issue; slide 10 gives earlier, market-targeted decisions

IMP3	Implementation, target users, organisational benefits	Three users, each with a benefit; monthly one-command update after arrivals are published
IMP4	Can expand, integrate, scale	Market view beside the national number (integration); more markets and states; forecast ranges next
IMP5	Fits the Datathon theme and SDGs	Theme named; SDG 8.9 and 12.b each with a reason; National Tourism Policy 2020–2030
INV1	Creative, non-routine approach	Market-by-market testing nobody had done; publish-date rule
INV2	Unique idea, clear identity	One message repeated: "market by market" (slides 6, 9, 12)
INV3	Engaging, professional, impactful	Opening question about 2022; A/B/C question paid off on slide 5; memorable last line
INV4	Wow factor	The 2022 story: risk nearly doubled yet arrivals jumped, shown live in the dashboard
INV5	Innovative, relevant use of AI/ML or automation	LightGBM across 20 markets; automated one-command monthly pipeline

Q&A answers (5 minutes)

Short answers in simple English. Backup slides: A accuracy, B markets, C data, D demo screenshots. **Who answers:** fill in names before the day. B and C take model questions, C takes data questions, D takes dashboard questions, A takes impact questions. Whoever answers, keep it to two or three sentences, then stop.

The finding: slopes and markets

What does a slope mean? B · backup B

It is how much expected arrivals change when global risk goes up, with everything else held the same. For Singapore, a one standard-deviation rise in global risk goes with about 38,600 more arrivals a month, holding oil, exchange rate, season and past arrivals constant. Per index point, that is about 815.

What do you mean by "control" or "holding constant"? B

All the variables go into the model together, so each one's coefficient shows its own effect with the others held the same. The risk slope is what is left after oil, exchange rate, season and past arrivals have taken their share. They are independent variables like risk; we call them controls because our question is about risk.

You use LightGBM. Why do the slopes come from Ridge? B

Each model does one job. LightGBM gives the most accurate forecast, but it is made of many decision trees, so it has no single number for the effect of risk. Ridge is a linear equation, so we can read one risk slope per market. Both use the same inputs and the same data.

The global risk index is the same for every country. How can each market have its own slope? B or C

We add global risk multiplied by each country, so each market gets its own slope, measured from how that market's arrivals moved with risk over time. Because it comes from timing, a market's own trend can line up with risk by chance. That is why we call these associations, not causes.

Do 16 and 3 really add up to plus 0.08? B

Not as a sum. The plus 0.08 is the raw link with nothing held constant. The 16 and 3 are slopes after controls. The slide shows the idea that opposite movements mix together in the national number.

Are the slopes statistically significant? B or C

Ridge does not give p-values, and we have not computed intervals yet. The direction is the same for 16 markets, but some slopes are small; India, for example, is close to zero. Bootstrap intervals are a clear next step.

Why do Singapore, Indonesia and China go up when risk goes up? A or B

We don't claim risk causes it. When wars like Ukraine and the Middle East push global risk up, long trips can feel less safe, so people in nearby markets may choose closer, familiar and cheaper trips, like Malaysia. At the same time, Malaysia made entry easier, with visa-free entry for China and the Visit Malaysia 2026 campaign. Our model cannot separate these, so we report it as a pattern.

Why is Thailand almost zero? A or B

Many Thai visitors cross the land border for short trips, shopping and family, so far-away events change little. The link is close to zero, with no clear pattern either way.

Do the results hold if you remove the COVID years? B or C

The slopes use all years since 2017. Our COVID stress test checks forecast accuracy, not the slopes. Refitting on recent years only is a next step we would take to confirm the three positive markets.

Which events drive the global risk index? B or C

In our data, the biggest jumps are early 2022, when Russia invaded Ukraine; October 2023, when the Gaza war began; and June 2025, the Israel–Iran war. As a yearly average, risk went from about 82 in 2021 to about 158 in 2022.

How important is geopolitical risk in the forecast? B

It is the 6th most used input. Past arrivals and season matter more, which makes sense. Risk matters more for which direction each market moves than for how many tourists come overall.

Forecasting and accuracy

Is LightGBM really better than Ridge? B · backup A

It had the lowest overall error, 15,969, and the lowest RMSE. But Ridge won 7 of the 20 test cells, so our choice is reasonable, not a big win.

Where does "7 percent better than repeating last month" come from? B · backup A

Repeating last month had an average error of 17,188 arrivals; LightGBM had 15,969. That is about 1,200 fewer, or about 7 percent lower.

Why not deep learning, like LSTM, or ARIMA? C

We have about 2,300 country-months, which is small for deep learning. One boosted-tree model trained on many short series together is a proven approach, as in the M5 forecasting competition. ARIMA fits one series at a time and cannot share patterns across markets.

Why do you report error in arrivals, not percent? B

Small markets have small numbers, so a percentage error for each market can jump a lot even when the miss is small in arrivals. Error in arrivals is stable and easy to read.

What is the 16.8% on the Forecast page? D

It is WAPE: total error divided by total arrivals, across all 20 markets. It is the same accuracy as our average error, shown as a share of arrivals.

How do you stop the model from seeing the future? C · backup C

Every input is delayed by its real publishing time. The model only learns from arrivals that were already published on the forecast date.

What if the arrivals data is published later or earlier than you assumed? C · backup A

We tested a 3-month and a 4-month delay. The best model stayed the same.

Data

Which of your data comes from DOSM? C

Our GDP and jobs figures come from DOSM's Tourism Satellite Account. Arrivals come from Tourism Malaysia with the Immigration Department. We also tested DOSM economic indicators, but dropped them because they are published too late to use in a live forecast.

Why only 20 markets? C

They are Malaysia's major source markets and supply most international arrivals. The method works the same way for more markets; that is an extension, not a redesign.

Why is the 2026 bar so low on Tourism Drivers? D

2026 only has January to May so far. Compared with the same five months of 2025, arrivals are up 0.9 percent.

Use and impact

Concretely, how could Tourism Malaysia use this? A

To decide where and when to promote. When global risk rises, the results show which markets have tended to see fewer arrivals and which have not, and the forecast gives one to two months of notice.

How would you measure SDG impact? A · backup C

Our report lists indicators, such as tourism jobs, wages and local SME participation for SDG 8.9, and energy, water, waste and traffic for SDG 12.b.

Can it run without your laptop? D

Yes. The Power BI file has its data built in, the data file is included, and the Python pipeline updates with one command. Code is on GitHub: github.com/william20808/nexora_dosm.

Backup D (slide 16) holds the demo screenshots if Power BI fails.